15.05.2025(m)

SET-A

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: January 2025– April 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2.5 Hrs.
Programme code: 105 Programme: Honors in Software and Mobile Application Development	Class: SY	Semester: IV (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department/Section/Center: COMP	
Course Code: 216H05C401	Name of the Course: Fundamental of Swift	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q.1	Attempt any **four**	
i)	Compare Structs and Classes in swift	20
ii)	Explain Mutating methods in swift with the help of an example.	05
iii)	Explain optional binding using `if let` and `guard let`. Provide examples.	05
iv)	Differentiate between `try?`, `try!`, and `try` in Swift error handling. Explain with the help on example/s.	05
v)	Explain the concept of delegation in Swift with a simple real-world example.	05
vi)	Explain the concept of computed properties in Swift with an example.	05
Q.2	Attempt any **one**	
i)	Write a swift program to perform following: a. Write a Swift closure named `add` that takes two `Int` values and returns their sum. b. Write a function named `performOperation` that takes two `Int` values and a closure that takes two `Int` and returns an `Int`. c. Call `performOperation` with two numbers and the `add` closure, and print the result. Write another closure named `multiply` that multiplies two `Int` values and call `performOperation` with it.	10 10
ii)	Write a swift program for Basic Device Control with following conditions: – Create a protocol named `Switchable` with a `isOn` property (Bool) and a `toggle()` function. – Create a struct `Light` that conforms to `Switchable`. – Implement the `toggle()` function in `Light` to flip the `isOn` property. – Create an instance of `Light` and demonstrate calling the `toggle()` function twice, printing the `isOn` status after each call.	10
Q.3	Attempt the following	
i)	Using UIKit in Storyboard, design a multi-screen app for a food order system: • **Screen 1:** A menu with 3 food items listed with `UILabels` and	20

	`UISwitch` for selection, plus a "Next" `UIButton`. • **Screen 2:** Show a summary of selected items using `UILabels` and a total using a `UITextField` (non-editable). • Use Auto Layout to ensure spacing and alignment are consistent. • Use Horizontal Stack View to display items. • Use a push segue for navigation and pass the selected values using `prepare(for:sender:)`.	